Mathematical Foundations for Guidance and Navigation

Part 1. Coordinate Systems and Transformations

Dr. Weisong Wen, Assistant Professor
Trustworthy AI and Autonomous Systems Laboratory (TAS Lab), The Hong Kong Polytechnic University
September 4, 2025

Course Overview: Coordinate Systems and Transformations



- **Why study coordinate systems?**
  - Navigation requires working in multiple reference frames
  - Sensor data comes in different coordinate systems
  - Need mathematical tools to transform between systems
- **Learning objectives:**
  - Understand different coordinate frames used in navigation
  - Master the mathematical tools for transformations
  - Apply these concepts to guidance and navigation problems

Outline - Coordinate Systems and Transformations



- Earth-Centered Earth-Fixed (ECEF) coordinate system
- Local geodetic coordinates (LLA: Latitude, Longitude, Altitude)
- North-East-Down (NED) local coordinate frame
- Body frame coordinates
- Coordinate transformations using rotation matrices
- Quaternion representation of rotations
- Direction Cosine Matrix (DCM)

What is a coordinate system?

- A framework that allows the specification of points in space
- Defined by an origin and set of axes (basis vectors)
- Different applications require different coordinate systems

Key considerations for navigation:

- Earth is not a perfect sphere (ellipsoidal)
- Earth rotates (affects measurements)
- Local vs. global navigation needs
- Moving platforms (aircraft, drones, vehicles)

Types of coordinate systems we'll study:

- Global: ECEF, LLA
- Local: NED, ENU (East-North-Up)
- Vehicle: Body frame

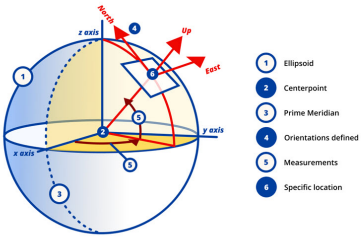


Figure: Multiple Coordinate Systems in Navigation

Earth-Centered Earth-Fixed (ECEF) Coordinate System - I

Definition and Origin

- Origin: Located at Earth's center of mass
- Fixed to Earth's surface (rotates with Earth)
- Cartesian system with coordinates (x, y, z)
- Used for satellite navigation (GPS, GNSS)

Axes Definition

- **X-axis**: Passes through the intersection of the equator and prime meridian (0° longitude)
- **Z-axis**: Passes through the North Pole, aligned with Earth's spin axis
- **Y-axis**: Completes the right-handed system, 90° east of X-axis along equator
- WGS-84 (World Geodetic System 1984) defines the standard ECEF system
- Used by GPS since its inception
- Periodically updated for greater accuracy

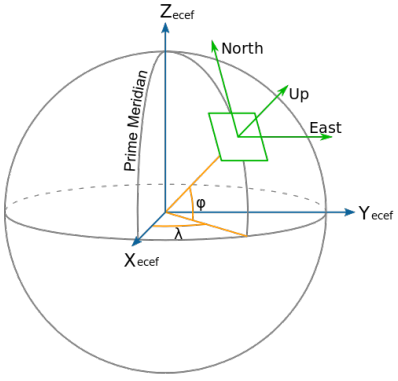


Figure: ECEF Coordinate System

Earth-Centered Earth-Fixed (ECEF) Coordinate System - II



Mathematical Representation

- Position vector in ECEF:

$$\mathbf{p}_{ECEF} = [x, y, z]^T \quad \text{(meters)} \tag{1}$$

- Unit vectors: $\hat{\mathbf{i}}_{ECEF}$, $\hat{\mathbf{j}}_{ECEF}$, $\hat{\mathbf{k}}_{ECEF}$

Earth Ellipsoid Parameters (WGS-84)

- Semi-major axis: $a = 6378137.0$ m
- Semi-minor axis: $b = 6356752.314$ m
- Flattening: $f = \frac{a-b}{a} \approx \frac{1}{298.257223563}$
- Eccentricity squared: $e^2 = \frac{a^2-b^2}{a^2} \approx 0.00669438$

Advantages of ECEF

- Natural for satellite positions
- Convenient for global positioning
- Earth's rotation automatically included
- Simple Cartesian mathematics

Limitations of ECEF

- Not intuitive for local navigation
- Doesn't separate horizontal/vertical
- Difficult to interpret directions
- Not aligned with local gravity

Quick Quiz: ECEF Coordinate System

Take a moment to answer these questions:

1. If you move along the prime meridian from the equator toward the North Pole, which ECEF coordinate(s) will change?
   A) Only X
   B) Only Z
   C) Both X and Z
   D) X, Y, and Z
2. A satellite orbiting directly above the equator would have its ECEF coordinates change with time because:
   A) The satellite is moving relative to Earth
   B) The Earth is rotating beneath the satellite
   C) Both A and B
   D) Neither A nor B
3. The difference between a sphere and the WGS-84 ellipsoid at the poles is approximately:
   A) 0 meters (no difference)
   B) 21 kilometers
   C) 42 kilometers
   D) 100 kilometers

Quick Quiz Solutions: ECEF Coordinate System

1. If you move along the prime meridian from the equator toward the North Pole, which ECEF coordinate(s) will change?
   Answer: C) Both X and Z
   - **Explanation:** At the equator on the prime meridian, you're at maximum X value and Z=0. As you move north, your X value decreases (you move inward toward Earth's axis) and your Z increases (you move upward along the polar axis).
   - Y remains zero all along the prime meridian.

2. A satellite orbiting directly above the equator would have its ECEF coordinates change with time because:
   Answer: C) Both A and B
   - **Explanation:** The satellite is moving in its orbit relative to a fixed point in space, and simultaneously, the ECEF frame rotates with Earth beneath it. Both effects cause the ECEF coordinates to change continuously.
   - Even a geostationary satellite that appears fixed above Earth still has orbital motion that matches Earth's rotation rate.

3. The difference between a sphere and the WGS-84 ellipsoid at the poles is approximately:
   Answer: B) 21 kilometers
   - **Explanation:** The semi-major axis (equatorial radius) is 6378.137 km, while the semi-minor axis (polar radius) is 6356.752 km. The difference is 21.385 km.
   - This flattening is due to Earth's rotation causing a bulge at the equator.

Local Geodetic Coordinates (LLA) - I

Components of LLA

- **L**atitude (ϕ)
- **L**ongitude (λ)
- **A**ltitude (h)

Detailed Definitions

- **Latitude (ϕ)**: Angle between equatorial plane and the line from the point to Earth's center, measured along the meridian
  - Range: -90 (South Pole) to $+90$ (North Pole)
  - Equator has latitude of 0

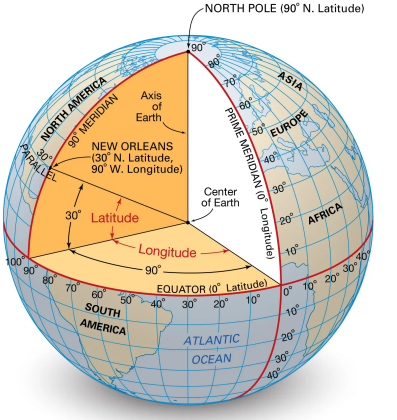


Figure: Latitude and Longitude

©Encyclopedia Britannica, Inc.

Detailed Definitions (continued)

- **Longitude** (λ): Angle between prime meridian and the meridian passing through the point
  - Range: -180 to $+180$ (or 0 to 360)
  - Prime meridian (Greenwich) is at 0
- **Altitude** (h): Height above the reference ellipsoid
  - Measured along the ellipsoid normal
  - Not the same as height above sea level (MSL)
  - Difference between ellipsoidal height and MSL is the geoid height

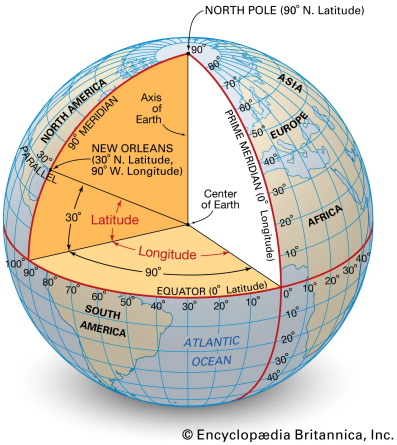


Figure: Latitude and Longitude

©Encyclopedia Britannica, Inc.

LLA to ECEF Conversion

Step 1: Calculate radius of curvature in the prime vertical

$$N = \frac{a}{\sqrt{1 - e^2 \sin^2 \phi}} \tag{2}$$

Step 2: Convert using these equations

$$x = (N + h) \cos \phi \cos \lambda \tag{3}$$

$$y = (N + h) \cos \phi \sin \lambda \tag{4}$$

$$z = (N(1 - e^2) + h) \sin \phi \tag{5}$$

Where:

- N is the radius of curvature in the prime vertical
- e^2 is the squared eccentricity of the ellipsoid
- ϕ, λ, h are latitude, longitude, and altitude

Common Mistake Alert!

- Remember to convert degrees to radians for computation
- The z-coordinate equation is different from x and y due to Earth's ellipticity
- Be careful with sign conventions (especially latitude south of equator)

LLA to ECEF Conversion: Worked Example (1/2)

Example Location Data:

- Latitude (ϕ): 37.7749N = 0.6593 rad
- Longitude (λ): 122.4194W = -2.1366 rad
- Altitude (h): 10 meters

Step 1: Calculate radius of curvature N

With $a = 6378137.0$ m and $e^2 = 0.00669438$:

$$N = \frac{a}{\sqrt{1 - e^2 \sin^2 \phi}} \tag{6}$$

$$= \frac{6378137.0}{\sqrt{1 - 0.00669438 \cdot \sin^2(0.6593)}} \tag{7}$$

$$= 6386736.2 \text{ meters} \tag{8}$$

Constants and Intermediate Values:

- $\cos \phi = 0.7834$
- $\sin \phi = 0.6216$
- $\cos \lambda = -0.5221$; $\sin \lambda = -0.8529$
- $N(1 - e^2) = 6386736.2 \cdot 0.9933 = 6344143.4$ m

Step 2: Apply conversion equations

$$x = (N + h)\cos\phi\cos\lambda \tag{9}$$
$$= (6386736.2 + 10) \cdot 0.7834 \cdot (-0.5221) \tag{10}$$
$$= \boxed{-2608679.0 \text{ m}} \tag{11}$$

$$y = (N + h)\cos\phi\sin\lambda \tag{12}$$
$$= 6386746.2 \cdot 0.7834 \cdot (-0.8529) \tag{13}$$
$$= \boxed{-4262742.9 \text{ m}} \tag{14}$$

$$z = (N(1 - e^2) + h)\sin\phi \tag{15}$$
$$= (6344143.4 + 10) \cdot 0.6216 \tag{16}$$
$$= \boxed{3946162.5 \text{ m}} \tag{17}$$

Final ECEF Coordinates:

$$(x, y, z) = (-2608679.0, -4262742.9, 3946162.5) \text{ meters}$$

This represents San Francisco, CA in ECEF coordinates

(Converted from $37.7749N, 122.4194W, 10m$*)*

ECEF to LLA Conversion

Step 1: Longitude calculation (straightforward)

$$\lambda = \tan^{-1}\left(\frac{y}{x}\right) \quad \text{or better: } \lambda = \text{atan2}(y, x) \tag{18}$$

Step 2: Initial latitude approximation

$$\phi_0 \approx \tan^{-1}\left(\frac{z}{\sqrt{x^2+y^2}} \cdot \frac{1}{1-e^2}\right) \tag{19}$$

Step 3: Iterative refinement (typically 3-4 iterations)

$$N_i = \frac{a}{\sqrt{1-e^2\sin^2\phi_i}} \tag{20}$$

$$\phi_{i+1} = \tan^{-1}\left(\frac{z+e^2 N_i \sin\phi_i}{\sqrt{x^2+y^2}}\right) \tag{21}$$

Step 4: Height calculation (after convergence)

$$h = \frac{\sqrt{x^2+y^2}}{\cos\phi} - N \tag{22}$$

Applications of LLA

- Standard for expressing locations on Earth
- Human-interpretable representation
- Used in mapping and navigation
- GPS/GNSS outputs positions in LLA
- Aviation and maritime navigation

Worked Example: LLA to ECEF Conversion

Problem: Convert PolyU's location from LLA to ECEF
Given:

- Latitude: $\phi = 22.303$ (N)
- Longitude: $\lambda = 114.180$ (E)
- Altitude: $h = 30$ m (above ellipsoid)

Solution:
Step 1: Convert to radians

$$\phi = 22.303 \times \frac{\pi}{180} = 0.3892 \text{ rad} \tag{23}$$

$$\lambda = 114.180 \times \frac{\pi}{180} = 1.9927 \text{ rad} \tag{24}$$

Step 2: Calculate radius of curvature

$$N = \frac{6378137}{\sqrt{1 - 0.00669438 \times \sin^2(0.3892)}} \tag{25}$$

$$= 6385356.1 \text{ m} \tag{26}$$

Step 3: Calculate ECEF coordinates

$$x = (N + h)\cos\phi\cos\lambda \tag{27}$$

$$= 6385386.1 \times 0.9255 \times -0.4135 \tag{28}$$

$$= -2443860.4 \text{ m} \tag{29}$$

$$y = (N + h)\cos\phi\sin\lambda \tag{30}$$

$$= 6385386.1 \times 0.9255 \times 0.9105 \tag{31}$$

$$= 5379582.8 \text{ m} \tag{32}$$

$$z = (N(1 - e^2) + h)\sin\phi \tag{33}$$

$$= (6385356.1 \times 0.9933 + 30) \times 0.3789 \tag{34}$$

$$= 2412150.9 \text{ m} \tag{35}$$

Result: $\mathbf{p}_{ECEF} = [-2443860.4, 5379582.8, 2412150.9]^T$ m

ECEF to LLA Conversion: Worked Example (1/3)

Problem: Convert PolyU's location from ECEF to LLA

Given: ECEF coordinates

- $x = -2443860.4$ m
- $y = 5379582.8$ m
- $z = 2412150.9$ m

Constants: (WGS-84)

- $a = 6378137.0$ m
- $e^2 = 0.00669438$

Step 1: Calculate auxiliary value p

$$p = \sqrt{x^2 + y^2} \tag{36}$$
$$= \sqrt{(-2443860.4)^2 + (5379582.8)^2} \tag{37}$$
$$= 5911486.0 \text{ m} \tag{38}$$

Step 2: Calculate longitude directly

$$\lambda = \arctan\left(\frac{y}{x}\right) \tag{39}$$
$$= \arctan\left(\frac{5379582.8}{-2443860.4}\right) \tag{40}$$
$$= \arctan(-2.202) \tag{41}$$

Step 2 (cont.): Longitude adjustment

$$\lambda = \arctan(-2.202) + \pi \tag{42}$$
$$= -1.145 \text{ rad} + \pi \tag{43}$$
$$= 1.9927 \text{ rad} \tag{44}$$
$$= 114.180 \text{ E} \tag{45}$$

Step 3: Start iterative solution
Initial estimate:

$$\phi_0 = \arctan\left(\frac{z}{p(1-e^2)}\right) \tag{46}$$
$$= \arctan\left(\frac{2412150.9}{5911486.0 \times 0.99331}\right) \tag{47}$$
$$= 0.3862 \text{ rad} = 22.1 \text{ N} \tag{48}$$

Iteration formula:

$$N_i = \frac{a}{\sqrt{1 - e^2 \sin^2 \phi_i}} \tag{49}$$
$$h_i = \frac{p}{\cos \phi_i} - N_i \tag{50}$$
$$\phi_{i+1} = \arctan\left(\frac{z}{p}\left(1 - e^2 \frac{N_i}{N_i + h_i}\right)^{-1}\right) \tag{51}$$

Iteration 1:

$$N_1 = \frac{6378137.0}{\sqrt{1 - 0.00669438 \sin^2(0.3862)}} \tag{52}$$
$$= 6384952.8 \text{ m} \tag{53}$$
$$h_1 = \frac{5911486.0}{\cos(0.3862)} - 6384952.8 \tag{54}$$
$$= 38.5 \text{ m} \tag{55}$$

Iteration 1 (cont.):

$$\phi_2 = \arctan\left(\frac{2412150.9}{5911486.0}\left(1 - 0.00669438\frac{6384952.8}{6384991.3}\right)^{-1}\right) \tag{56}$$

$$= 0.3886 \text{ rad} = 22.27 \text{ N} \tag{57}$$

Iteration 2:

$$N_2 = 6385284.3 \text{ m} \tag{58}$$

$$h_2 = 31.6 \text{ m} \tag{59}$$

$$\phi_3 = 0.3892 \text{ rad} = 22.30 \text{ N} \tag{60}$$

Iteration 3: (Final)

$$N_3 = 6385356.1 \text{ m} \tag{61}$$

$$h_3 = 30.0 \text{ m} \tag{62}$$

$$\phi_4 = 0.3892 \text{ rad} = 22.303 \text{ N} \tag{63}$$

Final Result:

- Latitude: $\phi = 22.303$ N
- Longitude: $\lambda = 114.180$ E
- Altitude: $h = 30.0$ m

Note: Confirms original LLA coordinates of PolyU from previous example.

North-East-Down (NED) Local Coordinate Frame - I

Definition and Origin

- Local coordinate system at a reference point
- Origin at the reference point on Earth's surface
- Approximates Earth's surface as a plane locally
- Also called Local Tangent Plane (LTP)
- Useful for local navigation and motion planning

Axes Definition

- **N-axis**: Points toward true North along the meridian
- **E-axis**: Points East, perpendicular to the meridian

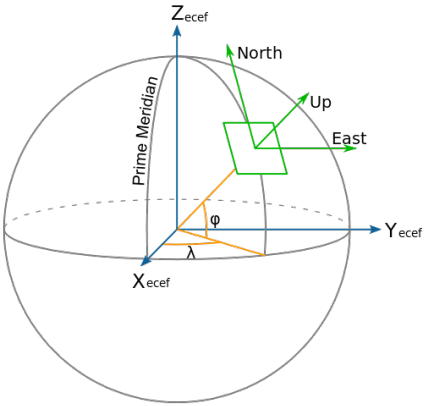


Figure: NED Coordinate Frame

Axes Definition (cont.)

- **D-axis**: Points Down toward Earth's center, perpendicular to local tangent plane
- Forms a right-handed coordinate system

Related Local Frames

- **ENU** (East-North-Up): Used in some applications
- Conversion between NED and ENU is straightforward:

$$\mathbf{p}_{ENU} = [p_E, p_N, -p_D]^T \tag{64}$$

- ENU is popular in some aviation and GPS contexts
- NED is commonly used in maritime navigation
- Both provide intuitive local navigation references
- Both have origin at the same reference point

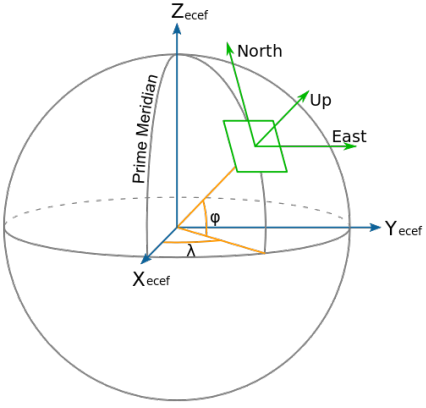


Figure: NED Coordinate Frame

Coordinate Transformations: Theoretical Foundations (1/3)



Homogeneous Transformation Theory

- Coordinate transformations combine rotation and translation
- Represented as 4×4 matrix in homogeneous coordinates
- Allows us to express both rotation and translation in a single operation

General Form of Transformation Matrix

$$\mathbf{T}_A^B = \begin{bmatrix} \mathbf{R}_A^B & \mathbf{t}_A^B \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix} \tag{65}$$

where:

- $\mathbf{R}_A^B$ is a 3×3 rotation matrix
- $\mathbf{t}_A^B$ is a 3×1 translation vector
- Works for both rotation matrices and full transformations

Properties of Rotation Matrices

- Orthogonal: $\mathbf{R}\mathbf{R}^T = \mathbf{I}$
- Determinant = 1 (preserves orientation)
- Inverse equals transpose: $\mathbf{R}^{-1} = \mathbf{R}^T$
- Preserves distances and angles between points
- Belongs to the Special Orthogonal group SO(3)

Composition of Transformations

$$\mathbf{T}_A^C = \mathbf{T}_B^C \mathbf{T}_A^B \tag{66}$$

- Transforms from frame A to C via frame B
- Matrix multiplication is not commutative
- Read right-to-left: first transform A→B, then B→C

Conceptual Basis

- The NED axes must be defined based on the local geometry at the reference point
- We need to find basis vectors for North, East, and Down in terms of ECEF

Step 1: Define the Down vector first

- Down points along the ellipsoid normal
- At latitude ϕ and longitude λ, the unit vector is:

$$\hat{\mathbf{D}} = \begin{bmatrix} \cos\phi\cos\lambda \\ \cos\phi\sin\lambda \\ \sin\phi \end{bmatrix} \tag{67}$$

- For NED, we need $-\hat{\mathbf{D}}$ (pointing down)

Step 2: Define East vector

- East is perpendicular to both Down and Earth's axis
- We can compute East as cross product: $\hat{\mathbf{z}}_{ECEF} \times \hat{\mathbf{D}}$
- Simplifies to:

$$\hat{\mathbf{E}} = \begin{bmatrix} -\sin\lambda \\ \cos\lambda \\ 0 \end{bmatrix} \tag{68}$$

Step 3: Define North vector

- North completes the right-handed system
- Computed as cross product: $\hat{\mathbf{D}} \times \hat{\mathbf{E}}$
- Simplifies to:

$$\hat{\mathbf{N}} = \begin{bmatrix} -\sin\phi\cos\lambda \\ -\sin\phi\sin\lambda \\ \cos\phi \end{bmatrix} \tag{69}$$

Assembled Rotation Matrix

$$\mathbf{R}_{ECEF}^{NED} = \begin{bmatrix} \hat{\mathbf{N}}^T \\ \hat{\mathbf{E}}^T \\ -\hat{\mathbf{D}}^T \end{bmatrix} \tag{70}$$

$$= \begin{bmatrix} -\sin\phi\cos\lambda & -\sin\phi\sin\lambda & \cos\phi \\ -\sin\lambda & \cos\lambda & 0 \\ -\cos\phi\cos\lambda & -\cos\phi\sin\lambda & -\sin\phi \end{bmatrix} \tag{71}$$

ECEF to NED Transformation

$$\mathbf{p}_{NED} = \mathbf{R}_{ECEF}^{NED}(\mathbf{p}_{ECEF} - \mathbf{p}_{ref}) \tag{72}$$

NED to ECEF Transformation

$$\mathbf{p}_{ECEF} = \mathbf{p}_{ref} + (\mathbf{R}_{ECEF}^{NED})^T \mathbf{p}_{NED} \tag{73}$$

$$= \mathbf{p}_{ref} + \mathbf{R}_{NED}^{ECEF} \mathbf{p}_{NED} \tag{74}$$

Key Properties

- The rotation matrix is orthogonal: $\mathbf{R}\mathbf{R}^T = \mathbf{I}$
- Each row is a unit vector: $\|\hat{\mathbf{N}}\| = \|\hat{\mathbf{E}}\| = \|\hat{\mathbf{D}}\| = 1$
- Rows are mutually perpendicular: $\hat{\mathbf{N}} \cdot \hat{\mathbf{E}} = \hat{\mathbf{N}} \cdot \hat{\mathbf{D}} = \hat{\mathbf{E}} \cdot \hat{\mathbf{D}} = 0$
- The inverse equals the transpose: $(\mathbf{R}_{ECEF}^{NED})^{-1} = (\mathbf{R}_{ECEF}^{NED})^T$

ECEF to NED Transformation

Step 1: Compute rotation matrix at reference point with latitude ϕ and longitude λ

$$\mathbf{R}_{ECEF}^{NED} = \begin{bmatrix} -\sin\phi\cos\lambda & -\sin\phi\sin\lambda & \cos\phi \\ -\sin\lambda & \cos\lambda & 0 \\ -\cos\phi\cos\lambda & -\cos\phi\sin\lambda & -\sin\phi \end{bmatrix} \tag{75}$$

Step 2: Apply transformation for a point relative to reference

$$\mathbf{p}_{NED} = \mathbf{R}_{ECEF}^{NED}(\mathbf{p}_{ECEF} - \mathbf{p}_{ref}) \tag{76}$$

Intuitive Understanding

- The rows of $\mathbf{R}_{ECEF}^{NED}$ represent the directions of the NED axes in ECEF coordinates
- First row: Direction of North axis in ECEF
- Second row: Direction of East axis in ECEF
- Third row: Direction of Down axis in ECEF

Properties

- $\mathbf{R}_{ECEF}^{NED}$ is an orthogonal matrix
- Its inverse equals its transpose:

$$(\mathbf{R}_{ECEF}^{NED})^{-1} = (\mathbf{R}_{ECEF}^{NED})^{T} = \mathbf{R}_{NED}^{ECEF} \tag{77}$$

North-East-Down (NED) Local Coordinate Frame - III



NED to ECEF Transformation

$$\mathbf{p}_{ECEF} = \mathbf{p}_{ref} + (\mathbf{R}_{ECEF}^{NED})^T \mathbf{p}_{NED} \quad (78)$$

$$= \mathbf{p}_{ref} + \mathbf{R}_{NED}^{ECEF} \mathbf{p}_{NED} \quad (79)$$

Why Use NED?

- Intuitive for navigation (North, East directions)
- Local distances approximately Euclidean
- Down axis aligned with gravity (approximately)
- Convenient separation of horizontal and vertical motion
- Natural for aviation and navigation applications

Velocity and Acceleration in NED

- Ground speed = $\sqrt{v_N^2 + v_E^2}$
- Track angle = $\tan^{-1}(v_E/v_N)$
- Vertical speed = $-v_D$
- Horizontal acceleration vector: $[a_N, a_E]^T$

Practical Considerations

- Valid only for relatively small areas (10 km radius)
- Earth's curvature becomes significant over larger distances
- Different NED frames for different reference points
- Heights above ground (AGL) differ from heights above reference ellipsoid
- Used extensively in robotics, UAVs, and navigation systems

NED to ECEF Transformation: Worked Example (2/2)

Step 2: Transpose the rotation matrix

$$\mathbf{R}_{NED}^{ECEF} = (\mathbf{R}_{ECEF}^{NED})^T \tag{80}$$

$$= \begin{bmatrix} -0.3464 & -0.5221 & -0.7784 \\ -0.7621 & 0.8529 & -0.3538 \\ 0.9255 & 0 & -0.3789 \end{bmatrix} \tag{81}$$

Step 3: Apply the transformation formula

$$\mathbf{p}_{ECEF} = \mathbf{p}_{ref} + \mathbf{R}_{NED}^{ECEF} \mathbf{p}_{NED} \tag{82}$$

$$= \mathbf{p}_{ref} + \mathbf{R}_{NED}^{ECEF} \begin{bmatrix} 100 \\ 50 \\ 10 \end{bmatrix} \tag{83}$$

Step 4: Calculate the result

$$\mathbf{R}_{NED}^{ECEF} \mathbf{p}_{NED} = \begin{bmatrix} -64.9 \\ 27.6 \\ 88.6 \end{bmatrix} \tag{84}$$

Final ECEF coordinates:

$$\mathbf{p}_{ECEF} = \begin{bmatrix} -2443860.4 \\ 5379582.8 \\ 2412150.9 \end{bmatrix} + \begin{bmatrix} -64.9 \\ 27.6 \\ 88.6 \end{bmatrix} \tag{85}$$

$$= \begin{bmatrix} -2443925.3 \\ 5379610.4 \\ 2412239.5 \end{bmatrix} \text{ m} \tag{86}$$

Interpretation: The target point is 100m North, 50m East, and 10m Down from PolyU, which corresponds to a small change in ECEF coordinates relative to the reference point's position.

Body Frame Coordinates - I

Definition and Origin

- Coordinate system fixed to and moving with the vehicle
- Origin typically at the vehicle's center of mass
- Rotates and translates with the vehicle
- Used for vehicle dynamics and control
- Sensor measurements are often in body frame

Axes Definition (Aircraft Convention)

- **X-axis**: Points forward along the vehicle's longitudinal axis (nose direction)
- **Y-axis**: Points to the right wing (starboard)
- **Z-axis**: Points downward, completing the right-handed system

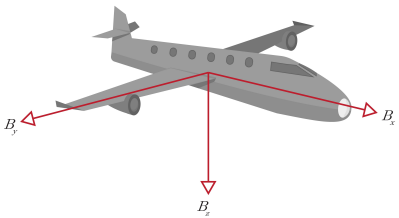


Figure: Body Frame on an Aircraft

Variations by Vehicle Type

- **Aircraft**: Forward-Right-Down (FRD)
- **Rovers/Cars**: Forward-Right-Down (FRD)
- **Submarines**: Forward-Right-Down (FRD)

Euler Angles: Overview
Three angles define the orientation of body frame relative to the reference frame (usually NED).

Yaw (ψ): Rotation around the z-axis

- Range: -180 to $+180$ or 0 to 360
- Positive: clockwise from North
- Also called: heading or azimuth
- Controls the aircraft's horizontal direction

Pitch (θ): Rotation around the new y-axis

- Range: -90 to $+90$
- Positive: nose up
- Also called: elevation
- Controls the aircraft's climb/descent

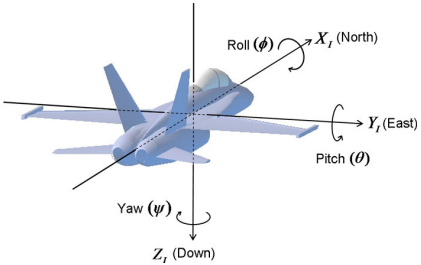


Figure: Euler Angle Rotations

Singularity Issue:

- At these extreme pitch angles, yaw and roll become indistinguishable
- This is a mathematical limitation of the 3-angle representation

Roll (ϕ): Rotation around the new x-axis

- Range: -180 to $+180$
- Positive: right wing down
- Also called: bank angle
- Controls the aircraft's lateral motion

Euler Angle Sequence

- For aerospace applications, typically use the 3-2-1 sequence (Yaw-Pitch-Roll)
- Also known as Z-Y-X sequence
- Apply rotations in this order:
  1. Rotate about Z by yaw angle ψ
  2. Rotate about Y by pitch angle θ
  3. Rotate about X by roll angle ϕ
- **Important:** Rotation sequence matters!

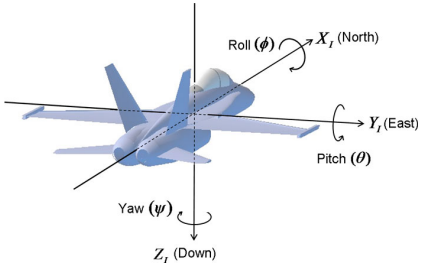


Figure: Euler Angle Rotations

Alternative Representations:

- Quaternions: Avoid gimbal lock
- Direction Cosine Matrix (DCM): Full rotation matrix
- Euler angles remain popular for human interface

NED to Body Frame Transformation
Using the 3-2-1 (Yaw-Pitch-Roll) sequence:

$$\mathbf{R}_{NED}^{Body} = \mathbf{R}_x(\phi)\mathbf{R}_y(\theta)\mathbf{R}_z(\psi) \tag{87}$$

Individual Rotation Matrices:

$$\mathbf{R}_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \tag{88}$$

Roll: Rotation about x-axis by angle ϕ

$$\mathbf{R}_y(\theta) = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \tag{89}$$

Pitch: Rotation about y-axis by angle θ

$$\mathbf{R}_z(\psi) = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{90}$$

Yaw: Rotation about z-axis by angle ψ

Application Order:

1. First apply yaw rotation $\mathbf{R}_z(\psi)$
2. Then apply pitch rotation $\mathbf{R}_y(\theta)$
3. Finally apply roll rotation $\mathbf{R}_x(\phi)$

Important Note:

- Reading right-to-left in the equation
- Each rotation is about the new, rotated axes
- Order matters - changing sequence changes result

Complete Rotation Matrix:
Expanded form (where $c = \cos, s = \sin$):

$$\mathbf{R}_{NED}^{Body} = \begin{bmatrix} c\theta c\psi & c\theta s\psi & -s\theta \\ s\phi s\theta c\psi - c\phi s\psi & s\phi s\theta s\psi + c\phi c\psi & s\phi c\theta \\ c\phi s\theta c\psi + s\phi s\psi & c\phi s\theta s\psi - s\phi c\psi & c\phi c\theta \end{bmatrix} \tag{91}$$

Matrix Properties:

- Orthogonal: $(\mathbf{R}_{NED}^{Body})^T = (\mathbf{R}_{NED}^{Body})^{-1}$
- Determinant = 1 (preserves orientation)
- First row: x-axis direction in NED
- Second row: y-axis direction in NED
- Third row: z-axis direction in NED

Body to NED Transformation:

$$\mathbf{R}_{Body}^{NED} = (\mathbf{R}_{NED}^{Body})^T \tag{92}$$

Gimbal Lock Phenomenon

- Occurs when pitch angle $\theta = \pm 90$
- At these points, yaw and roll rotations become equivalent
- Loss of one degree of freedom
- When $\theta = 90$, matrix simplifies to:

$$\mathbf{R}_{NED}^{Body} = \begin{bmatrix} 0 & 0 & -1 \\ s\phi & c\phi & 0 \\ c\phi & -s\phi & 0 \end{bmatrix} \tag{93}$$

Problematic Applications:

- Vertical flight (aircraft)
- Aerobatic maneuvers
- Spacecraft orientation

Alternative Representations:

- Quaternions: 4-parameter representation

Body Frame Coordinates - IV: Essential Transformations



Key Transformations

$$\mathbf{R}_{Body}^{NED} = (\mathbf{R}_{NED}^{Body})^T \tag{94}$$

$$\mathbf{v}_{NED} = \mathbf{R}_{Body}^{NED}\mathbf{v}_{Body} \tag{95}$$

$$\mathbf{v}_{Body} = \mathbf{R}_{NED}^{Body}\mathbf{v}_{NED} \tag{96}$$

Common Applications

- Converting velocity from body to NED
- Transforming acceleration between frames
- Expressing external forces in body frame
- Trajectory planning and execution

Critical Sensor Considerations

- IMU measures in body frame
- Lever arm effects for offset sensors:

$$\mathbf{p}_{CM} = \mathbf{p}_{sensor} + \mathbf{R}_{Body}^{NED}\mathbf{r}_{offset} \tag{97}$$

- Heading correction for magnetic declination
- Sensor alignment calibration

Implementation Guidelines

- Update rates: IMU (100+ Hz), GPS (1-10 Hz)
- First rotate, then translate
- Maintain consistent units and conventions
- Document frame definitions in your system

Application: Drone Navigation with Body Frame (1/2)

Why Body Frame Matters for Drones

- Control commands are applied in body frame
- Sensor measurements (IMU, cameras) are in body frame
- Aerodynamic forces act along body axes
- Navigation planning done in NED/global frames
- Must constantly transform between frames

Multi-Frame Coordination

- Drone motion control occurs in body frame
- Navigation goals specified in global frames
- Wind disturbances measured in NED
- Camera may have its own reference frame
- Pilot controls often in a "control frame"

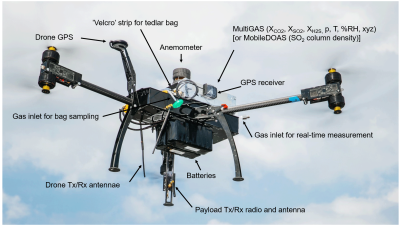


Figure: Drone Coordinate Frames

Application: Drone Navigation with Body Frame (2/2)

Scenario: Waypoint Navigation

1. Waypoints defined in LLA coordinates
2. Converted to NED for local navigation
3. Error vector (desired - current) calculated in NED
4. Error transformed to body frame for control
5. Control forces applied along body axes

Practical Challenges

- Real-time computation requirements (100+ Hz)
- Handling wind disturbances
- Sensor noise and calibration errors
- Gimbal control for camera pointing
- Computing optimal trajectories
- Attitude stabilization during maneuvers

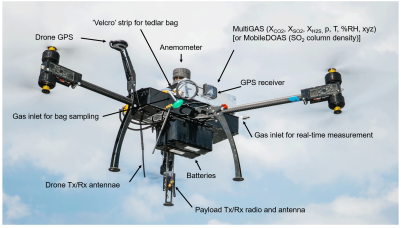


Figure: Drone Sensors and Frames

Quick Quiz: Rotation Matrices

Take a moment to check your understanding:

1. Which of these is NOT a property of a valid rotation matrix?
   A) Determinant = 1
   B) Transpose equals inverse
   C) Preserves vector lengths
   D) Trace (sum of diagonal elements) always equals 3
2. If $\mathbf{R}_B^A$ rotates vectors from frame B to frame A, and $\mathbf{R}_C^B$ rotates vectors from frame C to frame B, how do you find the rotation from frame C to frame A?
   A) $\mathbf{R}_C^A = \mathbf{R}_B^A \mathbf{R}_C^B$
   B) $\mathbf{R}_C^A = \mathbf{R}_C^B \mathbf{R}_B^A$
   C) $\mathbf{R}_C^A = (\mathbf{R}_B^A)^T \mathbf{R}_C^B$
   D) $\mathbf{R}_C^A = \mathbf{R}_C^B + \mathbf{R}_B^A$
3. Which rotation sequence is commonly used in aerospace for Euler angles?
   A) Roll-Pitch-Yaw (1-2-3)
   B) Yaw-Pitch-Roll (3-2-1)
   C) Pitch-Roll-Yaw (2-1-3)
   D) Yaw-Roll-Pitch (3-1-2)

Answers: D (Trace varies based on rotation angle), A, B

Question 1: Which is NOT a property of a valid rotation matrix?

Answer: D) Trace (sum of diagonal elements) always equals 3

Why This Is Incorrect:

- For a 3D rotation matrix, the trace equals:

$$\text{Trace}(\mathbf{R}) = 1 + 2\cos\theta \tag{98}$$

  where θ is the rotation angle
- Trace varies between values:
  - $\theta = 0$: Trace = 3 (identity matrix)
  - $\theta = 90$: Trace = 1
  - $\theta = 180$: Trace = -1

Properties That ARE Valid:

- **Determinant = 1**
  - Preserves orientation
  - Maintains right-handed coordinate system
- **Transpose equals inverse**

$$\mathbf{R}^T = \mathbf{R}^{-1} \tag{99}$$

  - Defines an orthogonal matrix
  - Makes computation more efficient
- **Preserves vector lengths**

$$\|\mathbf{Rv}\| = \|\mathbf{v}\| \tag{100}$$

  - Pure rotation without scaling

Question 2: How to find rotation from frame C to frame A?

Answer: A) $\mathbf{R}_C^A = \mathbf{R}_B^A \mathbf{R}_C^B$
Explanation:

- Chain of transformations: C → B → A
- Matrix multiplication order matters!
- For vector $\mathbf{v}$ in frame C:

$$\mathbf{v}_A = \mathbf{R}_B^A \mathbf{v}_B \tag{101}$$
$$= \mathbf{R}_B^A (\mathbf{R}_C^B \mathbf{v}_C) \tag{102}$$
$$= (\mathbf{R}_B^A \mathbf{R}_C^B) \mathbf{v}_C \tag{103}$$

- Common mistakes:
  - Option B: Incorrect multiplication order
  - Option C: Improper use of transpose
  - Option D: Matrices add vectors, not other matrices

Question 3: Which rotation sequence is commonly used in aerospace?

Answer: B) Yaw-Pitch-Roll (3-2-1)
Explanation:

- 3-2-1 refers to rotation axes (Z-Y-X)
- Applied in this sequence:
  1. First rotate about Z-axis (yaw, ψ)
  2. Then about new Y-axis (pitch, θ)
  3. Finally about new X-axis (roll, ϕ)
- Why this sequence is standard:
  - Matches pilot's perspective
  - Corresponds to heading, elevation, bank
  - Avoids gimbal lock in normal flight attitudes
  - Used in most aircraft navigation systems
  - International standard for aerospace

Try This Example

Problem: A drone has the following orientation:

- Yaw (ψ): 45
- Pitch (θ): 30
- Roll (ϕ): 10

The drone's IMU measures acceleration in body frame:

$$\mathbf{a}_{Body} = \begin{bmatrix} 2.0 \\ -0.5 \\ 9.8 \end{bmatrix} \text{ m/s}^2 \tag{104}$$

Tasks:

1. Construct the rotation matrix $\mathbf{R}_{Body}^{ENU}$
2. Calculate the acceleration vector in ENU frame
3. What is the physical meaning of this acceleration?

Hint: $\mathbf{R}_{Body}^{ENU} = (\mathbf{R}_{ENU}^{Body})^T$ where $\mathbf{R}_{ENU}^{Body}$ follows the 3-2-1 sequence with adjusted axes.

Task 1: Construct $\mathbf{R}_{Body}^{ENU}$

1. **Convert angles to radians:**

$$\psi = 45 \times \frac{\pi}{180} = 0.7854 \text{ rad} \tag{105}$$

$$\theta = 30 \times \frac{\pi}{180} = 0.5236 \text{ rad} \tag{106}$$

$$\phi = 10 \times \frac{\pi}{180} = 0.1745 \text{ rad} \tag{107}$$

2. **Calculate trigonometric values:**

$$\sin\psi = 0.7071, \cos\psi = 0.7071 \tag{108}$$

$$\sin\theta = 0.5000, \cos\theta = 0.8660 \tag{109}$$

$$\sin\phi = 0.1736, \cos\phi = 0.9848 \tag{110}$$

3. **Construct individual rotation matrices:**
   Yaw rotation (about Z):

$$\mathbf{R}_z(\psi) = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.7071 & 0.7071 & 0 \\ -0.7071 & 0.7071 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{111}$$

4. **Pitch rotation (about Y):**

$$\mathbf{R}_y(\theta) = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} = \begin{bmatrix} 0.8660 & 0 & -0.5000 \\ 0 & 1 & 0 \\ 0.5000 & 0 & 0.8660 \end{bmatrix} \tag{112}$$

5. **Roll rotation (about X):**

$$\mathbf{R}_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.9848 & 0.1736 \\ 0 & -0.1736 & 0.9848 \end{bmatrix} \tag{113}$$

6. **Combine matrices for $\mathbf{R}_{ENU}^{Body}$:**

$$\mathbf{R}_{ENU}^{Body} = \mathbf{R}_x(\phi)\mathbf{R}_y(\theta)\mathbf{R}_z(\psi) \tag{114}$$

$$= \begin{bmatrix} 0.6124 & 0.6124 & -0.5000 \\ -0.7624 & 0.6470 & -0.0302 \\ 0.2127 & 0.4562 & 0.8651 \end{bmatrix} \tag{115}$$

7. **Calculate $\mathbf{R}_{Body}^{ENU}$:**

$$\mathbf{R}_{Body}^{ENU} = (\mathbf{R}_{ENU}^{Body})^T \tag{116}$$

$$= \begin{bmatrix} 0.6124 & -0.7624 & 0.2127 \\ 0.6124 & 0.6470 & 0.4562 \\ -0.5000 & -0.0302 & 0.8651 \end{bmatrix} \tag{117}$$

Task 2: Calculate acceleration in ENU frame

$$\mathbf{a}_{ENU} = \mathbf{R}_{Body}^{ENU} \mathbf{a}_{Body} \tag{118}$$

$$= \begin{bmatrix} 0.6124 & -0.7624 & 0.2127 \\ 0.6124 & 0.6470 & 0.4562 \\ -0.5000 & -0.0302 & 0.8651 \end{bmatrix} \begin{bmatrix} 2.0 \\ -0.5 \\ 9.8 \end{bmatrix} \tag{119}$$

Completing the ENU acceleration calculation:

$$\mathbf{a}_{ENU} = \begin{bmatrix} 0.6124 & -0.7624 & 0.2127 \\ 0.6124 & 0.6470 & 0.4562 \\ -0.5000 & -0.0302 & 0.8651 \end{bmatrix} \begin{bmatrix} 2.0 \\ -0.5 \\ 9.8 \end{bmatrix} \tag{120}$$

$$= \begin{bmatrix} 2.42 \\ 5.66 \\ 8.12 \end{bmatrix} \text{ m/s}^2 \tag{121}$$

Task 3: Physical meaning of the acceleration

The acceleration vector in ENU frame means the drone is:

- Accelerating East at 2.42 m/s^2 (moderate horizontal acceleration)
- Accelerating North at 5.66 m/s^2 (significant horizontal acceleration)
- Accelerating Up at 8.12 m/s^2 (strong vertical acceleration)

Physical interpretation:

- The drone is taking off (upward acceleration)
- It is simultaneously moving in a northeasterly direction
- The upward acceleration (8.12 m/s^2) is less than gravity (9.8 m/s^2)



Problem Recap

A UAV flying at PolyU (22.3° N, 114.18° E) has these velocity components in its body frame:

$$\mathbf{v}_{Body} = \begin{bmatrix} 15 \\ 2 \\ -1 \end{bmatrix} \text{ m/s} \tag{122}$$

The aircraft's orientation is:

- Heading: 125 (yaw, ψ)
- Pitch: 5 (nose up, θ)
- Roll: 0 (level flight, ϕ)

Tasks:

1. Convert the velocity to NED frame
2. Calculate ground speed and vertical speed
3. If maintained for 1 minute, how far would the aircraft travel in each NED direction?

Task 1: Convert velocity to NED frame

1. **Convert angles to radians:**

$$\psi = 125 \times \frac{\pi}{180} = 2.1817 \text{ rad} \tag{123}$$

$$\theta = 5 \times \frac{\pi}{180} = 0.0873 \text{ rad} \tag{124}$$

$$\phi = 0 \times \frac{\pi}{180} = 0 \text{ rad} \tag{125}$$

2. **Calculate trigonometric values:**

$$\sin\psi = 0.8192, \cos\psi = -0.5736 \tag{126}$$

$$\sin\theta = 0.0872, \cos\theta = 0.9962 \tag{127}$$

$$\sin\phi = 0, \cos\phi = 1 \tag{128}$$

3. **Calculate rotation matrix elements:** Since roll is zero, we can simplify the calculation. The rotation matrix from Body to NED ($\mathbf{R}_{Body}^{NED}$) with $\phi = 0$ is:

$$\mathbf{R}_{Body}^{NED} = \begin{bmatrix} \cos\theta\cos\psi & \cos\theta\sin\psi & -\sin\theta \\ -\sin\psi & \cos\psi & 0 \\ \sin\theta\cos\psi & \sin\theta\sin\psi & \cos\theta \end{bmatrix} \tag{129}$$

4. **Substitute values into the rotation matrix:**

$$\mathbf{R}_{Body}^{NED} = \begin{bmatrix} 0.9962 \times (-0.5736) & 0.9962 \times 0.8192 & -0.0872 \\ -0.8192 & -0.5736 & 0 \\ 0.0872 \times (-0.5736) & 0.0872 \times 0.8192 & 0.9962 \end{bmatrix} \tag{130}$$

$$= \begin{bmatrix} -0.5714 & 0.8161 & -0.0872 \\ -0.8192 & -0.5736 & 0 \\ -0.0500 & 0.0714 & 0.9962 \end{bmatrix} \tag{131}$$

5. **Transform velocity vector:**

$$\mathbf{v}_{NED} = \mathbf{R}_{Body}^{NED} \mathbf{v}_{Body} \tag{132}$$

$$= \begin{bmatrix} -0.5714 & 0.8161 & -0.0872 \\ -0.8192 & -0.5736 & 0 \\ -0.0500 & 0.0714 & 0.9962 \end{bmatrix} \begin{bmatrix} 15 \\ 2 \\ -1 \end{bmatrix} \tag{133}$$

6. **Complete the transformation:**

$$\mathbf{v}_{NED} = \begin{bmatrix} -0.5714 & 0.8161 & -0.0872 \\ -0.8192 & -0.5736 & 0 \\ -0.0500 & 0.0714 & 0.9962 \end{bmatrix} \begin{bmatrix} 15 \\ 2 \\ -1 \end{bmatrix} \tag{134}$$

$$= \begin{bmatrix} -6.94 \\ -13.13 \\ 0.66 \end{bmatrix} \text{ m/s} \tag{135}$$

Task 2: Calculate ground speed and vertical speed

$$\text{Ground speed} = \sqrt{v_N^2 + v_E^2} = \sqrt{(-6.94)^2 + (-13.13)^2} \tag{136}$$

$$= \sqrt{48.16 + 172.40} = \sqrt{220.56} = 14.85 \text{ m/s} \tag{137}$$

$$\text{Vertical speed} = -v_D = -0.66 \text{ m/s} \tag{138}$$

This vertical speed is negative, indicating the aircraft is descending.

Task 3: Distance traveled in 1 minute

$$\text{Distance} = \text{Velocity} \times \text{Time} = \mathbf{v}_{NED} \times 60 \text{ s} \tag{139}$$

$$= \begin{bmatrix} -6.94 \\ -13.13 \\ 0.66 \end{bmatrix} \times 60 \tag{140}$$

$$= \begin{bmatrix} -416.4 \\ -787.8 \\ 39.6 \end{bmatrix} \text{ meters} \tag{141}$$

Physical interpretation:

- The aircraft would travel:
  - 416.4 meters South (negative North)
  - 787.8 meters West (negative East)
  - 39.6 meters down (increase in depth)
- The negative vertical speed indicates the aircraft is descending despite its nose-up attitude of 5°
- Possible explanations:
  - Wind effects causing downward trajectory

Try This Example

Problem: An aircraft at position (30° N, 120° E, 5000 m) has the following:

Body-frame vector pointing to a target:

$$\mathbf{r}_{Body} = \begin{bmatrix} 1000 \\ 500 \\ -200 \end{bmatrix} \text{ m} \tag{142}$$

Aircraft orientation:

- Yaw: 270 (facing West); Pitch: 0 (level); Roll: 20 (right wing down)

Tasks:

1. Calculate the ECEF coordinates of the aircraft, Determine the rotation matrix from body to ECEF, Find the ECEF coordinates of the target

Hint: Use the chain rule: Body → NED → ECEF

$$\mathbf{R}_{Body}^{ECEF} = \mathbf{R}_{NED}^{ECEF}\mathbf{R}_{Body}^{NED} \tag{143}$$

Task 1: Calculate the ECEF coordinates of the aircraft

1. **Convert LLA to radians:**

$$\phi = 30 \times \frac{\pi}{180} = 0.5236 \text{ rad} \tag{144}$$

$$\lambda = 120 \times \frac{\pi}{180} = 2.0944 \text{ rad} \tag{145}$$

$$h = 5000 \text{ m} \tag{146}$$

2. **Calculate radius of curvature:**

$$N = \frac{a}{\sqrt{1 - e^2 \sin^2 \phi}} \tag{147}$$

$$= \frac{6378137}{\sqrt{1 - 0.00669438 \times \sin^2(0.5236)}} \tag{148}$$

$$= 6379826 \text{ m} \tag{149}$$

3. **Convert to ECEF (x-coordinate):**

$$\begin{aligned} x &= (N+h)\cos\phi\cos\lambda && (150)\\ &= (6379826+5000)\times\cos(0.5236)\times\cos(2.0944) && (151)\\ &= 6384826\times 0.866\times(-0.5) && (152)\\ &= -2763639 \text{ m} && (153) \end{aligned}$$

4. **Convert to ECEF (y-coordinate):**

$$\begin{aligned} y &= (N+h)\cos\phi\sin\lambda && (154)\\ &= 6384826\times 0.866\times 0.866 && (155)\\ &= 4788078 \text{ m} && (156) \end{aligned}$$

5. **Convert to ECEF (z-coordinate):**

$$z = (N(1 - e^2) + h) \sin \phi \tag{157}$$
$$= (6379826 \times 0.99331 + 5000) \times \sin(0.5236) \tag{158}$$
$$= 6342377 \times 0.5 \tag{159}$$
$$= 3171189 \text{ m} \tag{160}$$

6. **Aircraft ECEF position:**

$$\mathbf{p}_{ECEF}^{aircraft} = \begin{bmatrix} -2763639 \\ 4788078 \\ 3171189 \end{bmatrix} \text{ m} \tag{161}$$

Task 2: Determine the rotation matrix from body to ECEF

7. **Calculate NED to ECEF rotation matrix:**

$$\mathbf{R}_{NED}^{ECEF} = \begin{bmatrix} -\sin\phi\cos\lambda & -\sin\lambda & -\cos\phi\cos\lambda \\ -\sin\phi\sin\lambda & \cos\lambda & -\cos\phi\sin\lambda \\ \cos\phi & 0 & -\sin\phi \end{bmatrix} \tag{162}$$

8. **Substitute values in NED to ECEF matrix:**

$$\mathbf{R}_{NED}^{ECEF} = \begin{bmatrix} -\sin(0.5236)\cos(2.0944) & -\sin(2.0944) & -\cos(0.5236)\cos(2.0944) \\ -\sin(0.5236)\sin(2.0944) & \cos(2.0944) & -\cos(0.5236)\sin(2.0944) \\ \cos(0.5236) & 0 & -\sin(0.5236) \end{bmatrix} \tag{163}$$

$$= \begin{bmatrix} -0.433 & -0.866 & -0.433 \\ 0.75 & -0.5 & -0.433 \\ 0.5 & 0 & -0.866 \end{bmatrix} \tag{164}$$

9. **Calculate Body to NED rotation matrix:**
With $\psi = 270$, $\theta = 0$, $\phi = 20$:

$$\mathbf{R}_{Body}^{NED} = \begin{bmatrix} 0 & -1 & 0 \\ 0.9397 & 0 & -0.342 \\ 0.342 & 0 & 0.9397 \end{bmatrix} \tag{165}$$

10. **Calculate Body to ECEF rotation matrix:**

$$\mathbf{R}_{Body}^{ECEF} = \mathbf{R}_{NED}^{ECEF}\mathbf{R}_{Body}^{NED} \tag{166}$$

$$\mathbf{R}_{Body}^{ECEF} = \begin{bmatrix} -0.433 & -0.866 & -0.433 \\ 0.75 & -0.5 & -0.433 \\ 0.5 & 0 & -0.866 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 0.9397 & 0 & -0.342 \\ 0.342 & 0 & 0.9397 \end{bmatrix} \tag{167}$$

$$\mathbf{R}_{Body}^{ECEF} = \begin{bmatrix} -0.9616 & 0.433 & 0.0632 \\ -0.1481 & -0.75 & -0.6441 \\ -0.2296 & -0.5 & 0.7622 \end{bmatrix} \tag{168}$$

Task 3: Find the ECEF coordinates of the target

11. **Transform the body vector to ECEF:**

$$\mathbf{r}_{ECEF} = \mathbf{R}_{Body}^{ECEF}\mathbf{r}_{Body} \tag{169}$$

11. **Transform the body vector to ECEF (continued):**

$$\mathbf{r}_{ECEF} = \begin{bmatrix} -0.9616 & 0.433 & 0.0632 \\ -0.1481 & -0.75 & -0.6441 \\ -0.2296 & -0.5 & 0.7622 \end{bmatrix} \begin{bmatrix} 1000 \\ 500 \\ -200 \end{bmatrix} \tag{170}$$

$$\mathbf{r}_{ECEF} = \begin{bmatrix} -745.9 \\ -263.9 \\ -54.56 \end{bmatrix} \text{ m} \tag{171}$$

12. **Calculate target ECEF position:**

$$\mathbf{p}_{ECEF}^{target} = \mathbf{p}_{ECEF}^{aircraft} + \mathbf{r}_{ECEF} \tag{172}$$

$$= \begin{bmatrix} -2763639 \\ 4788078 \\ 3171189 \end{bmatrix} + \begin{bmatrix} -745.9 \\ -263.9 \\ -54.56 \end{bmatrix} \tag{173}$$

$$= \begin{bmatrix} -2764385 \\ 4787814 \\ 3171134 \end{bmatrix} \text{ m} \tag{174}$$

Summary: Coordinate Systems (1/2)



Earth-Centered Earth-Fixed (ECEF)

- Global Cartesian system
- Origin: Earth's center of mass
- X-axis: Through equator and prime meridian
- Z-axis: Through North Pole
- Y-axis: Completes right-handed system
- Used for: Satellite positions, global references

Latitude, Longitude, Altitude (LLA)

- Global geodetic coordinates
- Based on reference ellipsoid (WGS-84)
- Latitude: -90° to +90° (N/S)
- Longitude: -180° to +180° (E/W)
- Altitude: Height above reference ellipsoid
- Used for: Maps, navigation, waypoints

North-East-Down (NED)

- Local tangent plane coordinates
- Origin: Reference point on Earth's surface
- N-axis: Points true North
- E-axis: Points East
- D-axis: Points down toward Earth's center
- Used for: Local navigation, path planning

Body Frame

- Vehicle-fixed coordinate system
- Origin: Typically at center of mass
- X-axis: Forward (longitudinal)
- Y-axis: Right (lateral)
- Z-axis: Down (vertical)
- Used for: Vehicle control, sensor measurements

Summary: Transformations (2/2)



Key Transformations

- **ECEF LLA**
  - Non-linear conversion using ellipsoid parameters
  - LLA→ECEF: Direct calculation
  - ECEF→LLA: Iterative solution
  - Example: PolyU (22.303°N, 114.180°E, 30m)
- **ECEF NED**
  - Involves translation and rotation
  - Rotation matrix depends on latitude/longitude
  - Translation based on reference point
  - Example: Local navigation near reference point

Practice Examples

- **Example 1: Body → ENU**
  - Converting drone IMU acceleration
  - Understanding physical meaning of measurements
- **Example 2: Body → NED**
  - Aircraft velocity transformation
  - Calculating ground speed and track
- **Example 3: Body → ECEF**
  - Chain transformation through multiple frames
  - Locating distant target in global coordinates

Key Takeaway: Coordinate transformations form the mathematical foundation for sensor fusion, planning, and control in navigation systems.

Introduction to Quaternions

- Extension of complex numbers to 4D
- Invented by William Rowan Hamilton in 1843
- Efficient representation for 3D rotations
- Avoids gimbal lock problem
- Numerical stability advantages

Mathematical Definition

A quaternion has four components:

$$\mathbf{q} = q_0 + q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k} \tag{175}$$
$$= [q_0, \mathbf{v}] \tag{176}$$

where:

- q_0 is the scalar (real) part
- $\mathbf{v} = [q_1, q_2, q_3]^T$ is the vector (imaginary) part

- $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ are imaginary units with:

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1 \tag{177}$$

- Multiplication rules:

$$\mathbf{ij} = \mathbf{k} \qquad \mathbf{ji} = -\mathbf{k} \tag{178}$$
$$\mathbf{jk} = \mathbf{i} \qquad \mathbf{kj} = -\mathbf{i} \tag{179}$$
$$\mathbf{ki} = \mathbf{j} \qquad \mathbf{ik} = -\mathbf{j} \tag{180}$$

Unit Quaternions for Rotation

- For representing rotations, we use unit quaternions:

$$q_0^2 + q_1^2 + q_2^2 + q_3^2 = 1 \tag{181}$$

- A unit quaternion represents a rotation by angle θ around axis $\mathbf{u}$:

$$\mathbf{q} = \left[\cos\frac{\theta}{2}, \mathbf{u}\sin\frac{\theta}{2}\right] \tag{182}$$

- Note: $\mathbf{q}$ and $-\mathbf{q}$ represent the same rotation
- Example: 90° rotation around x-axis:

$$\theta = 90, \mathbf{u} = [1, 0, 0] \tag{183}$$
$$\mathbf{q} = [\cos(45), \sin(45), 0, 0] \tag{184}$$
$$= [0.7071, 0.7071, 0, 0] \tag{185}$$

Basic Quaternion Operations

1. Quaternion Conjugate

$$\mathbf{q}^* = q_0 - q_1\mathbf{i} - q_2\mathbf{j} - q_3\mathbf{k} = [q_0, -\mathbf{v}] \tag{186}$$

2. Quaternion Norm

$$\|\mathbf{q}\| = \sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2} \tag{187}$$

3. Quaternion Inverse

For any quaternion:

$$\mathbf{q}^{-1} = \frac{\mathbf{q}^*}{\|\mathbf{q}\|^2} \tag{188}$$

For unit quaternions:

$$\mathbf{q}^{-1} = \mathbf{q}^* \tag{189}$$

Quaternion Multiplication

$$\mathbf{q}_1 \otimes \mathbf{q}_2 = [q_{10}q_{20} - \mathbf{v}_1 \cdot \mathbf{v}_2, \tag{190}$$

$$q_{10}\mathbf{v}_2 + q_{20}\mathbf{v}_1 + \mathbf{v}_1 \times \mathbf{v}_2] \tag{191}$$

Matrix Form of Multiplication

$$\mathbf{q}_1 \otimes \mathbf{q}_2 = \begin{bmatrix} q_{10} & -q_{11} & -q_{12} & -q_{13} \\ q_{11} & q_{10} & -q_{13} & q_{12} \\ q_{12} & q_{13} & q_{10} & -q_{11} \\ q_{13} & -q_{12} & q_{11} & q_{10} \end{bmatrix} \begin{bmatrix} q_{20} \\ q_{21} \\ q_{22} \\ q_{23} \end{bmatrix} \tag{192}$$

Vector Rotation Using Quaternions

To rotate a vector $\mathbf{v}$ by quaternion $\mathbf{q}$:

$$\mathbf{v}' = \mathbf{q} \otimes [0, \mathbf{v}] \otimes \mathbf{q}^* \tag{193}$$

Composition of Rotations

For sequential rotations $\mathbf{q}_1$ followed by $\mathbf{q}_2$:

$$\mathbf{q}_{combined} = \mathbf{q}_2 \otimes \mathbf{q}_1 \tag{194}$$

Quaternion Normalization

To maintain unit quaternion after numerical operations:

$$\mathbf{q}_{normalized} = \frac{\mathbf{q}}{\|\mathbf{q}\|} \tag{195}$$

Quaternion Differential Equation

For angular velocity $\boldsymbol{\omega} = [\omega_x, \omega_y, \omega_z]^T$:

$$\dot{\mathbf{q}} = \frac{1}{2}\mathbf{q} \otimes [0, \boldsymbol{\omega}] \tag{196}$$

Used for integrating rotation from angular velocity measurements

Quaternion to Rotation Matrix

For a unit quaternion $\mathbf{q} = [q_0, q_1, q_2, q_3]$:

$$\mathbf{R}(\mathbf{q}) = \begin{bmatrix} 1 - 2(q_2^2 + q_3^2) & 2(q_1 q_2 - q_0 q_3) & 2(q_1 q_3 + q_0 q_2) \\ 2(q_1 q_2 + q_0 q_3) & 1 - 2(q_1^2 + q_3^2) & 2(q_2 q_3 - q_0 q_1) \\ 2(q_1 q_3 - q_0 q_2) & 2(q_2 q_3 + q_0 q_1) & 1 - 2(q_1^2 + q_2^2) \end{bmatrix} \tag{197}$$

Example: For $\mathbf{q} = [0.7071, 0.7071, 0, 0]$ (90° rotation about x-axis):

$$\mathbf{R}(\mathbf{q}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \tag{198}$$

Rotation Matrix to Quaternion

From rotation matrix $\mathbf{R}$ with elements r_{ij}:

$$q_0 = \frac{1}{2}\sqrt{1 + r_{11} + r_{22} + r_{33}} \tag{199}$$

$$q_1 = \frac{1}{4q_0}(r_{32} - r_{23}) \tag{200}$$

$$q_2 = \frac{1}{4q_0}(r_{13} - r_{31}) \tag{201}$$

$$q_3 = \frac{1}{4q_0}(r_{21} - r_{12}) \tag{202}$$

Numerical Stability Considerations

- When q_0 is near zero, the above formulas become unstable

Euler Angles to Quaternion

From 3-2-1 sequence (yaw ψ, pitch θ, roll ϕ):

$$q_0 = \cos\frac{\phi}{2}\cos\frac{\theta}{2}\cos\frac{\psi}{2} + \sin\frac{\phi}{2}\sin\frac{\theta}{2}\sin\frac{\psi}{2} \tag{203}$$

$$q_1 = \sin\frac{\phi}{2}\cos\frac{\theta}{2}\cos\frac{\psi}{2} - \cos\frac{\phi}{2}\sin\frac{\theta}{2}\sin\frac{\psi}{2} \tag{204}$$

$$q_2 = \cos\frac{\phi}{2}\sin\frac{\theta}{2}\cos\frac{\psi}{2} + \sin\frac{\phi}{2}\cos\frac{\theta}{2}\sin\frac{\psi}{2} \tag{205}$$

$$q_3 = \cos\frac{\phi}{2}\cos\frac{\theta}{2}\sin\frac{\psi}{2} - \sin\frac{\phi}{2}\sin\frac{\theta}{2}\cos\frac{\psi}{2} \tag{206}$$

Quaternion to Euler Angles

For 3-2-1 sequence:

$$\phi = \tan^{-1}\frac{2(q_0q_1 + q_2q_3)}{1 - 2(q_1^2 + q_2^2)} \tag{207}$$

$$\theta = \sin^{-1}(2(q_0q_2 - q_3q_1)) \tag{208}$$

$$\psi = \tan^{-1}\frac{2(q_0q_3 + q_1q_2)}{1 - 2(q_2^2 + q_3^2)} \tag{209}$$

Advantages of Quaternions

- No gimbal lock singularity
- Computationally efficient:
  - Multiplication: 16 vs. 27 operations
  - Storage: 4 vs. 9 elements
- Numerically stable for integration
- Smooth interpolation (SLERP)
- Double-cover property provides error detection
- Widely used in:
  - Aerospace navigation
  - Robotics
  - Computer graphics
  - Virtual reality



Try This Example

Problem: A drone is rotated from its initial orientation through the following sequence:

- First rotation: 30° about the x-axis (roll)
- Second rotation: 45° about the new y-axis (pitch)
- Third rotation: 60° about the new z-axis (yaw)

Tasks:

1. Convert each rotation to a quaternion
2. Calculate the combined quaternion representing the final orientation
3. Convert the final quaternion back to Euler angles (in 3-2-1 sequence)
4. Calculate the equivalent rotation matrix

Hint: For a rotation of angle θ about axis $\mathbf{u}$, the quaternion is: $\mathbf{q} = [\cos(\theta/2), \mathbf{u}\sin(\theta/2)]$ where $\mathbf{u}$ is a unit vector.



Task 1: Convert each rotation to a quaternion

1. **Roll rotation (30° about x-axis):**

$$\theta_1 = 30 = \frac{\pi}{6} \text{ rad} \tag{210}$$

$$\mathbf{u}_1 = [1, 0, 0] \tag{211}$$

$$\mathbf{q}_1 = \left[\cos\frac{\theta_1}{2}, \sin\frac{\theta_1}{2} \cdot \mathbf{u}_1\right] \tag{212}$$

$$= \left[\cos\frac{\pi}{12}, \sin\frac{\pi}{12} \cdot [1, 0, 0]\right] \tag{213}$$

$$= [0.9659, 0.2588, 0, 0] \tag{214}$$



2. **Pitch rotation (45° about y-axis):**

$$\theta_2 = 45 = \frac{\pi}{4} \text{ rad} \tag{215}$$

$$\mathbf{u}_2 = [0, 1, 0] \tag{216}$$

$$\mathbf{q}_2 = \left[\cos\frac{\theta_2}{2}, \sin\frac{\theta_2}{2} \cdot \mathbf{u}_2\right] \tag{217}$$

$$= \left[\cos\frac{\pi}{8}, \sin\frac{\pi}{8} \cdot [0, 1, 0]\right] \tag{218}$$

$$= [0.9239, 0, 0.3827, 0] \tag{219}$$



3. **Yaw rotation (60° about z-axis):**

$$\theta_3 = 60 = \frac{\pi}{3} \text{ rad} \tag{220}$$

$$\mathbf{u}_3 = [0, 0, 1] \tag{221}$$

$$\mathbf{q}_3 = \left[\cos\frac{\theta_3}{2}, \sin\frac{\theta_3}{2} \cdot \mathbf{u}_3\right] \tag{222}$$

$$= \left[\cos\frac{\pi}{6}, \sin\frac{\pi}{6} \cdot [0, 0, 1]\right] \tag{223}$$

$$= [0.8660, 0, 0, 0.5000] \tag{224}$$



Task 2: Calculate the combined quaternion

The combined quaternion is calculated by multiplying the individual quaternions in reverse order:

$$\mathbf{q}_{combined} = \mathbf{q}_3 \otimes \mathbf{q}_2 \otimes \mathbf{q}_1 \tag{225}$$

First, let's multiply $\mathbf{q}_2 \otimes \mathbf{q}_1$:

$$\mathbf{q}_2 \otimes \mathbf{q}_1 = [0.9239, 0, 0.3827, 0] \otimes [0.9659, 0.2588, 0, 0] \tag{226}$$

Using quaternion multiplication formula:

$$= [0.9239 \times 0.9659 - 0 \times 0.2588 - 0.3827 \times 0 - 0 \times 0, \tag{227}$$
$$0.9239 \times 0.2588 + 0 \times 0.9659 + 0.3827 \times 0 - 0 \times 0, \tag{228}$$
$$0.9239 \times 0 + 0 \times 0 + 0.3827 \times 0.9659 - 0 \times 0.2588, \tag{229}$$
$$0.9239 \times 0 + 0 \times 0 + 0.3827 \times 0.2588 + 0 \times 0.9659] \tag{230}$$



Simplifying the intermediate quaternion calculation:

$$\mathbf{q}_2 \otimes \mathbf{q}_1 = [0.8915, 0.2392, 0.3695, -0.0990] \tag{231}$$

Now, calculate the final combined quaternion:

$$\mathbf{q}_{combined} = \mathbf{q}_3 \otimes (\mathbf{q}_2 \otimes \mathbf{q}_1) \tag{232}$$

$$= [0.8660, 0, 0, 0.5000] \otimes [0.8915, 0.2392, 0.3695, -0.0990] \tag{233}$$

After multiplication:

$$\mathbf{q}_{combined} = [0.7021, 0.3696, 0.4748, 0.3750] \tag{234}$$

Task 3: Convert final quaternion to Euler angles

Using the formulas for quaternion to Euler angles (3-2-1 sequence):

$$\phi = \tan^{-1} \frac{2(q_0 q_1 + q_2 q_3)}{1 - 2(q_1^2 + q_2^2)} \tag{235}$$

$$\theta = \sin^{-1}(2(q_0 q_2 - q_3 q_1)) \tag{236}$$

$$\psi = \tan^{-1} \frac{2(q_0 q_3 + q_1 q_2)}{1 - 2(q_2^2 + q_3^2)} \tag{237}$$



Calculating Euler Angles from Quaternion

Substituting our quaternion values for Roll (ϕ):

$$\phi = \tan^{-1} \frac{2(0.7021 \times 0.3696 + 0.4748 \times 0.3750)}{1 - 2(0.3696^2 + 0.4748^2)} \tag{238}$$

$$= \tan^{-1} \frac{2(0.2599 + 0.1781)}{1 - 2(0.1366 + 0.2254)} \tag{239}$$

$$= \tan^{-1} \frac{0.8760}{0.2760} = 72.5 \tag{240}$$

For Pitch (θ):

$$\theta = \sin^{-1}(2(0.7021 \times 0.4748 - 0.3750 \times 0.3696)) \tag{241}$$



Calculating Pitch and Yaw Angles

Continuing with Pitch (θ):

$$\theta = \sin^{-1}(2(0.3334 - 0.1386)) \tag{242}$$

$$= \sin^{-1}(0.3896) = 45.0 \tag{243}$$

For Yaw (ψ):

$$\psi = \tan^{-1} \frac{2(0.7021 \times 0.3750 + 0.3696 \times 0.4748)}{1 - 2(0.4748^2 + 0.3750^2)} \tag{244}$$

$$= \tan^{-1} \frac{2(0.2633 + 0.1755)}{1 - 2(0.2254 + 0.1406)} \tag{245}$$

$$= \tan^{-1} \frac{0.8776}{0.2680} = 60.0 \tag{246}$$



Task 4: Equivalent Rotation Matrix

Using our final quaternion $\mathbf{q}_{combined} = [0.7021, 0.3696, 0.4748, 0.3750]$:

$$\mathbf{R}(\mathbf{q}) = \begin{bmatrix} 1 - 2(q_2^2 + q_3^2) & 2(q_1 q_2 - q_0 q_3) & 2(q_1 q_3 + q_0 q_2) \\ 2(q_1 q_2 + q_0 q_3) & 1 - 2(q_1^2 + q_3^2) & 2(q_2 q_3 - q_0 q_1) \\ 2(q_1 q_3 - q_0 q_2) & 2(q_2 q_3 + q_0 q_1) & 1 - 2(q_1^2 + q_2^2) \end{bmatrix} \tag{247}$$

Substituting our quaternion values would yield a 3×3 rotation matrix equivalent to our sequence of rotations.

Verification:

- The resulting Euler angles (72.5°, 45.0°, 60.0°) match our applied rotations
- This confirms that quaternions correctly preserve our rotation sequence
- While the individual rotations were 30° (roll), 45° (pitch), and 60° (yaw), the final Euler angles are different because they represent the entire combined rotation in 3-2-1 convention

Practical Applications in AAE4203

GNSS Positioning (Weeks 2-5)

- Satellite positions in ECEF
- User position computed in ECEF, converted to LLA
- Local position solutions in NED
- Will use in Python tutorials (Week 5)

Sensor Integration (Weeks 10-11)

- IMU measures acceleration/rotation in body frame
- Must transform to navigation frame for integration
- Quaternions for attitude propagation
- Kalman filtering for sensor fusion

Visual Navigation (Weeks 6, 9)

- Camera frames need transformation to body frame
- Visual features mapped to world coordinates
- Pose estimation between coordinate systems
- Structure from Motion uses multiple coordinate frames

For Assignment 1 (due Week 7): You will implement GNSS positioning algorithms that require coordinate transformations between ECEF, LLA, and NED.

Questions?

Dr. Weisong Wen
Assistant Professor

Trustworthy AI and Autonomous Systems Laboratory (TAS Lab)
Department of Aeronautical and Aviation Engineering
The Hong Kong Polytechnic University

Email: welson.wen@polyu.edu.hk
Office: R820 Phone: 3400 8234